Exploratory Data Analysis (EDA) is the foundation of every data science project. It is the process of examining datasets to understand their structure, identify patterns, detect anomalies and extract meaningful insights. Before applying any machine learning or statistical models, data must be cleaned, transformed and explored this is where EDA plays an important role.

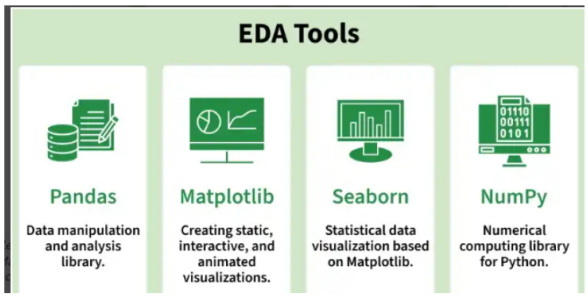
EDA helps answer important questions such as:

What type of data is present (numerical, categorical, text, dates)

Are there missing or inconsistent values

Are there outliers that could affect analysis

What patterns or relationships exist between variables



Data Cleaning vs Data Preprocessing

Feature	Data Cleaning	Data Preprocessing
Meaning	Fixing errors in raw data	Preparing data for ML models
Purpose	Improve data quality	Improve model performance
Focus	Missing values, duplicates, noise	Scaling, encoding, splitting, feature creation
Stage	Early stage	After cleaning + transformations
Scope	Narrow	Broad (includes cleaning)

👉 **Simple idea:**

- Data Cleaning = *Make data correct*
- Data Preprocessing = *Make data usable for machine learning*

Major Steps comnes in the Data Cleaning

These steps improve data accuracy:

- Remove duplicate rows
- Handle missing values (NaN)
- Fix wrong formats
- Correct spelling or noise
- Remove irrelevant columns
- Handle outliers

Major stpes in the Data Preprocessing?

This includes cleaning **plus** extra steps needed before training models:

- Data cleaning
- Encoding categorical variables
- Feature scaling (Normalization/Standardization)
- Feature engineering
- Train-test split
- Text preprocessing (tokenization, stemming)

👉 **Data Cleaning** → *Make data correct*

👉 **Data Wrangling** → *Make data structured & meaningful*
👉 **Data Preprocessing** → *Make data ready for machine learning*

🧹 **Data Cleaning**

- Remove empty **statement**
- Fix wrong dates
- Remove duplicate records

🔧 **Data Wrangling**

- Combine `author + source` into new feature
- Create text length feature
- Merge multiple datasets (PolitiFact + ISOT)

⚙️ **Data Preprocessing**

- Tokenization
- Stopword removal
- TF-IDF / n-gram features
- Label encoding (`BinaryNumTarget`)
- Train–test split

Double-click (or enter) to edit

How to Perform EDA Using Python?

Step 1: Import Python Libraries

Step 2: Reading Dataset

Step 3: Data Reduction

Step 4: Feature Engineering

Step 5: Creating Features

Step 6: Data Cleaning/Wrangling

Step 7: EDA Exploratory Data Analysis

Step 8: Statistics Summary

Step 9: EDA Univariate Analysis

Step 10: Data Transformation

Step 12: EDA Bivariate Analysis

Step 13: EDA Multivariate Analysis

Step 14: Impute Missing values

Exploratory Data Analysis in Python

Double-click (or enter) to edit

```
Start coding or generate with AI.
```

Import Python Libraries

Reading Dataset from Github

```
import pandas as pd
data = 'https://raw.githubusercontent.com/makhan010385/DataSet/refs/heads/main/used_cars_data.csv'
```

Analyzing the Data

Before we make any inferences, we listen to our data by examining all variables in the data.

The main goal of data understanding is to gain general insights about the data, which covers the number of rows and columns, values in the data, datatypes, and Missing values in the dataset.

shape – shape will display the number of observations(rows) and features(columns) in the dataset

There are 7253 observations and 14 variables in our dataset

head() will display the top 5 observations of the dataset

```
data = pd.read_csv(data)
print(data.head())
```

```
   S.No.                              Name    Location  Year  \
0      0           Maruti Wagon R LXI CNG      Mumbai  2010
1      1  Hyundai Creta 1.6 CRDi SX Option        Pune  2015
2      2                Honda Jazz V    Chennai  2011
3      3             Maruti Ertiga VDI     Chennai  2012
4      4   Audi A4 New 2.0 TDI Multitronic  Coimbatore  2013

   Kilometers_Driven Fuel_Type Transmission Owner_Type     Mileage   Engine  \
0              72000       CNG       Manual      First  26.6 km/kg   998 CC
1              41000    Diesel       Manual      First  19.67 kmpl  1582 CC
2              46000    Petrol       Manual      First   18.2 kmpl  1199 CC
3              87000    Diesel       Manual      First  20.77 kmpl  1248 CC
4              40670    Diesel    Automatic     Second   15.2 kmpl  1968 CC

       Power  Seats  New_Price  Price
0  58.16 bhp    5.0        NaN   1.75
1  126.2 bhp    5.0        NaN  12.50
2   88.7 bhp    5.0  8.61 Lakh   4.50
3  88.76 bhp    7.0        NaN   6.00
4  140.8 bhp    5.0        NaN  17.74
```

tail() will display the last 5 observations of the dataset

```
data.tail()
```

	S.No.	Name	Location	Year	Kilometers_Driven	Fuel_Type	Transmission	Owner_Type	Mileage	Engine	Power	Seats	New_Price	Price
7248	7248	Volkswagen Vento Diesel Trendline	Hyderabad	2011	89411	Diesel	Manual	First	20.54 kmpl	1598 CC	103.6 bhp	5.0	NaN	NaN
7249	7249	Volkswagen Polo GT TSI	Mumbai	2015	59000	Petrol	Automatic	First	17.21 kmpl	1197 CC	103.6 bhp	5.0	NaN	NaN
7250	7250	Nissan Micra Diesel XV	Kolkata	2012	28000	Diesel	Manual	First	23.08 kmpl	1461 CC	63.1 bhp	5.0	NaN	NaN
7251	7251	Volkswagen Polo GT TSI	Pune	2013	52262	Petrol	Automatic	Third	17.2 kmpl	1197 CC	103.6 bhp	5.0	NaN	NaN
7252	7252	Mercedes-Benz E-Class 2009-2013 E 220 CDI Avan...	Kochi	2014	72443	Diesel	Automatic	First	10.0 kmpl	2148 CC	170 bhp	5.0	NaN	NaN

data.info() shows the variables Mileage, Engine, Power, Seats, New_Price, and Price have missing values.

Numeric variables like Mileage, Power are of datatype as float64 and int64. Categorical variables like Location, Fuel_Type, Transmission, and Owner Type are of object data type.

Check for Duplication

nunique() based on several unique values in each column and the data description, we can identify the continuous and categorical columns in the data.

Duplicated data can be handled or removed based on further analysis

```
data.nunique()
```

	0
S.No.	7253
Name	2041
Location	11
Year	23
Kilometers_Driven	3660
Fuel_Type	5
Transmission	2
Owner_Type	4
Mileage	450
Engine	150
Power	386
Seats	9
New_Price	625
Price	1373

dtype: int64

Missing Values Calculation

isnull() is widely been in all pre-processing steps to identify null values in the data

In our example, data.isnull().sum() is used to get the number of missing records in each column

```
data.isnull().sum()
```

	0
S.No.	0
Name	0

Location	0
Year	0
Kilometers_Driven	0
Fuel_Type	0
Transmission	0
Owner_Type	0
Mileage	2
Engine	46
Power	46
Seats	53
New_Price	6247
Price	1234

dtype: int64

The below code helps to calculate the percentage of missing values in each column

```
(data.isnull().sum()/(len(data)))*100
```

	0
S.No.	0.000000
Name	0.000000
Location	0.000000
Year	0.000000
Kilometers_Driven	0.000000
Fuel_Type	0.000000
Transmission	0.000000
Owner_Type	0.000000
Mileage	0.027575
Engine	0.634220
Power	0.634220
Seats	0.730732
New_Price	86.129877
Price	17.013650

dtype: float64

The percentage of missing values for the columns New_Price and Price is ~86% and ~17%, respectively.

Step 3: Data Reduction

Some columns or variables can be dropped if they do not add value to our analysis.

In our dataset, the column S.No have only ID values, assuming they don't have any predictive power to predict the dependent variable.

```
# Remove S.No. column from data
data = data.drop(['S.No.'], axis = 1)
data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 7253 entries, 0 to 7252
Data columns (total 13 columns):
 #   Column             Non-Null Count  Dtype  
---  ------             --------------  -----  
 0   Name               7253 non-null   object 
 1   Location           7253 non-null   object 
 2   Year               7253 non-null   int64  
 3   Kilometers_Driven  7253 non-null   int64  
 4   Fuel_Type          7253 non-null   object 
 5   Transmission       7253 non-null   object 
 6   Owner_Type         7253 non-null   object 
 7   Mileage            7251 non-null   object 
 8   Engine             7207 non-null   object 
 9   Power              7207 non-null   object 
 10  Seats              7200 non-null   float64
 11  New_Price          1006 non-null   object 
 12  Price              6019 non-null   float64
dtypes: float64(2), int64(2), object(9)
memory usage: 736.8+ KB
```

We start our Feature Engineering as we need to add some columns required for analysis.

Step 4: Feature Engineering

Feature engineering refers to the process of using domain knowledge to select and transform the most relevant variables from raw data when creating a predictive model using machine

learning or statistical modeling.

The main goal of Feature engineering is to create meaningful data from raw data.

Step 5: Creating Features

We can play around with the variables Year and Name in our dataset. If we see the sample data, the column "Year" shows the manufacturing year of the car.

It would be difficult to find the car's age if it is in year format as the Age of the car is a contributing factor to Car Price.

Introducing a new column, "Car_Age" to know the age of the car

```
from datetime import date
date.today().year
data['Car_Age']=date.today().year-data['Year']
data.head()
```

	Name	Location	Year	Kilometers_Driven	Fuel_Type	Transmission	Owner_Type	Mileage	Engine	Power	Seats	New_Price	Price	Car_Age
0	Maruti Wagon R LXI CNG	Mumbai	2010	72000	CNG	Manual	First	26.6 km/kg	998 CC	58.16 bhp	5.0	NaN	1.75	16
1	Hyundai Creta 1.6 CRDi SX Option	Pune	2015	41000	Diesel	Manual	First	19.67 kmpl	1582 CC	126.2 bhp	5.0	NaN	12.50	11
2	Honda Jazz V	Chennai	2011	46000	Petrol	Manual	First	18.2 kmpl	1199 CC	88.7 bhp	5.0	8.61 Lakh	4.50	15
3	Maruti Ertiga VDI	Chennai	2012	87000	Diesel	Manual	First	20.77 kmpl	1248 CC	88.76 bhp	7.0	NaN	6.00	14
4	Audi A4 New 2.0 TDI Multitronic	Coimbatore	2013	40670	Diesel	Automatic	Second	15.2 kmpl	1968 CC	140.8 bhp	5.0	NaN	17.74	13

Since car names will not be great predictors of the price in our current data. But we can process this column to extract important information using brand and Model names.

Let's split the name and introduce new variables "Brand" and "Model"

```
data['Brand'] = data.Name.str.split().str.get(0)
```

```
data['Model'] = data.Name.str.split().str.get(1) + data.Name.str.split().str.get(2)
```

```
data[['Name','Brand','Model']]
```

	Name	Brand	Model
0	Maruti Wagon R LXI CNG	Maruti	WagonR
1	Hyundai Creta 1.6 CRDi SX Option	Hyundai	Creta1.6
2	Honda Jazz V	Honda	JazzV
3	Maruti Ertiga VDI	Maruti	ErtigaVDI
4	Audi A4 New 2.0 TDI Multitronic	Audi	A4New
...	...	...	...
7248	Volkswagen Vento Diesel Trendline	Volkswagen	VentoDiesel
7249	Volkswagen Polo GT TSI	Volkswagen	PoloGT
7250	Nissan Micra Diesel XV	Nissan	MicraDiesel
7251	Volkswagen Polo GT TSI	Volkswagen	PoloGT
7252	Mercedes-Benz E-Class 2009-2013 E 220 CDI Avan...	Mercedes-Benz	E-Class2009-2013

7253 rows × 3 columns

Step 6: Data Cleaning/Wrangling

Some names of the variables are not relevant and not easy to understand. Some data may have data entry errors, and some variables may need data type conversion. We need to fix this issue in the data.

In the example, The brand name 'Isuzu' 'ISUZU' and 'Mini' and 'Land' looks incorrect. This needs to be corrected

```
print(data.Brand.unique())
print(data.Brand.nunique())
```

```
['Maruti' 'Hyundai' 'Honda' 'Audi' 'Nissan' 'Toyota' 'Volkswagen' 'Tata'
 'Land' 'Mitsubishi' 'Renault' 'Mercedes-Benz' 'BMW' 'Mahindra' 'Ford'
 'Porsche' 'Datsun' 'Jaguar' 'Volvo' 'Chevrolet' 'Skoda' 'Mini' 'Fiat'
 'Jeep' 'Smart' 'Ambassador' 'Isuzu' 'ISUZU' 'Force' 'Bentley'
 'Lamborghini' 'Hindustan' 'OpelCorsa']
33
```

1 Basic Statistical Exploration

```
# Summary statistics
data.describe()
# Include categorical columns also
```

	Year	Kilometers_Driven	Seats	Price	Car_Age
count	7253.000000	7.253000e+03	7200.000000	6019.000000	7253.000000
mean	2013.365366	5.869906e+04	5.279722	9.479468	12.634634
std	3.254421	8.442772e+04	0.811660	11.187917	3.254421
min	1996.000000	1.710000e+02	0.000000	0.440000	7.000000
25%	2011.000000	3.400000e+04	5.000000	3.500000	10.000000
50%	2014.000000	5.341600e+04	5.000000	5.640000	12.000000
75%	2016.000000	7.300000e+04	5.000000	9.950000	15.000000
max	2019.000000	6.500000e+06	10.000000	160.000000	30.000000

Double-click (or enter) to edit

2 Value Counts Analysis (Very Important for Beginners)

```
# Check most common brands
data['Brand'].value_counts()
```

Brand	count
Maruti	1444
Hyundai	1340
Honda	743
Toyota	507
Mercedes-Benz	380
Volkswagen	374
Ford	351
Mahindra	331
BMW	312
Audi	285
Tata	228
Skoda	202
Renault	170
Chevrolet	151
Nissan	117
Land Rover	67
Jaguar	48
Fiat	38
Mitsubishi	36
Mini Cooper	31
Volvo	28
Porsche	19
Jeep	19
Datsun	17
Isuzu	5
Force	3
Bentley	2
Ambassador	1
Smart	1
Lamborghini	1

Data Type Separation

```
# Numeric columns
num_cols = data.select_dtypes(include=['int64','float64'])

# Categorical columns
cat_cols = data.select_dtypes(include=['object'])

print(num_cols.columns)
print(cat_cols.columns)
```

```
Index(['Year', 'Kilometers_Driven', 'Seats', 'Price', 'Car_Age'], dtype='object')
Index(['Name', 'Location', 'Fuel_Type', 'Transmission', 'Owner_Type',
       'Mileage', 'Engine', 'Power', 'New_Price', 'Brand', 'Model'],
      dtype='object')
```

3. Correlation Analysis

```
# Correlation matrix
corr_matrix = num_cols.corr()
corr_matrix
```

	Year	Kilometers_Driven	Seats	Price	Car_Age
Year	1.000000	-0.187859	0.008216	0.305327	-1.000000
Kilometers_Driven	-0.187859	1.000000	0.090221	-0.011493	0.187859
Seats	0.008216	0.090221	1.000000	0.052225	-0.008216
Price	0.305327	-0.011493	0.052225	1.000000	-0.305327
Car_Age	-1.000000	0.187859	-0.008216	-0.305327	1.000000

Positive vs Negative correlation

Strong vs weak relation

The values range from:

Value	Meaning
+1	Perfect positive relationship
0	No relationship
–1	Perfect negative relationship

How to Read Your Output

Year, Kilometers_Driven, Seats, Price, Car_Age

Each cell tells:

If one variable increases, what happens to the other?

1 Year vs Car_Age = -1.000000

Perfect negative correlation.

2 Year vs Price = 0.305327 (Moderate Positive)

Newer cars tend to have higher price.

But not very strong → other factors also matter.

3. Car_Age vs Price = -0.305327

Older cars → lower price.

This is logical in real life.

5 Duplicate Data Checking

```
# Check duplicates
data.duplicated().sum()
print(data)
```

```
                                          Name    Location  Year  \
0                        Maruti Wagon R LXI CNG      Mumbai  2010
1            Hyundai Creta 1.6 CRDi SX Option        Pune  2015
2                                  Honda Jazz V     Chennai  2011
3                             Maruti Ertiga VDI     Chennai  2012
4             Audi A4 New 2.0 TDI Multitronic  Coimbatore  2013
...                                        ...         ...   ...
7248        Volkswagen Vento Diesel Trendline   Hyderabad  2011
7249                  Volkswagen Polo GT TSI      Mumbai  2015
7250                  Nissan Micra Diesel XV     Kolkata  2012
7251                  Volkswagen Polo GT TSI        Pune  2013
7252  Mercedes-Benz E-Class 2009-2013 E 220 CDI Avan...       Kochi  2014

      Kilometers_Driven Fuel_Type Transmission Owner_Type     Mileage  \
0                 72000       CNG       Manual      First  26.6 km/kg
1                 41000    Diesel       Manual      First  19.67 kmpl
2                 46000    Petrol       Manual      First   18.2 kmpl
3                 87000    Diesel       Manual      First  20.77 kmpl
4                 40670    Diesel    Automatic     Second   15.2 kmpl
...                 ...       ...          ...        ...        ...
7248              89411    Diesel       Manual      First  20.54 kmpl
7249              59000    Petrol    Automatic      First  17.21 kmpl
7250              28000    Diesel       Manual      First  23.08 kmpl
7251              52262    Petrol    Automatic      Third   17.2 kmpl
7252              72443    Diesel    Automatic      First   10.0 kmpl
```

```
      Engine      Power  Seats  New_Price  Price  Car_Age          Brand  \
0     998 CC  58.16 bhp    5.0        NaN   1.75       16         Maruti
1    1582 CC  126.2 bhp    5.0        NaN  12.50       11        Hyundai
2    1199 CC   88.7 bhp    5.0  8.61 Lakh   4.50       15          Honda
3    1248 CC  88.76 bhp    7.0        NaN   6.00       14         Maruti
4    1968 CC  140.8 bhp    5.0        NaN  17.74       13           Audi
...      ...        ...    ...        ...    ...      ...            ...
7248 1598 CC  103.6 bhp    5.0        NaN    NaN       15     Volkswagen
7249 1197 CC  103.6 bhp    5.0        NaN    NaN       11     Volkswagen
7250 1461 CC   63.1 bhp    5.0        NaN    NaN       14         Nissan
7251 1197 CC  103.6 bhp    5.0        NaN    NaN       13     Volkswagen
7252 2148 CC    170 bhp    5.0        NaN    NaN       12  Mercedes-Benz

               Model
0             WagonR
1           Creta1.6
2              JazzV
3          ErtigaVDI
4               A4New
...               ...
7248      VentoDiesel
7249           PoloGT
7250      MicraDiesel
7251           PoloGT
7252  E-Class2009-2013

[7252 rows x 16 columns]
```

[ ]

```
# Remove duplicates
data = data.drop_duplicates()
print(data)
```

```
                                           Name    Location  Year  \
0                          Maruti Wagon R LXI CNG      Mumbai  2010
1                    Hyundai Creta 1.6 CRDi SX Option        Pune  2015
2                                 Honda Jazz V     Chennai  2011
3                            Maruti Ertiga VDI     Chennai  2012
4                     Audi A4 New 2.0 TDI Multitronic  Coimbatore  2013
...                                          ...         ...   ...
7248                Volkswagen Vento Diesel Trendline   Hyderabad  2011
7249                          Volkswagen Polo GT TSI      Mumbai  2015
7250                          Nissan Micra Diesel XV     Kolkata  2012
7251                          Volkswagen Polo GT TSI        Pune  2013
7252  Mercedes-Benz E-Class 2009-2013 E 220 CDI Avan...       Kochi  2014

      Kilometers_Driven Fuel_Type Transmission Owner_Type     Mileage  \
0                 72000       CNG       Manual      First  26.6 km/kg
1                 41000    Diesel       Manual      First  19.67 kmpl
2                 46000    Petrol       Manual      First   18.2 kmpl
3                 87000    Diesel       Manual      First  20.77 kmpl
4                 40670    Diesel    Automatic     Second   15.2 kmpl
...                 ...       ...          ...        ...         ...
7248              89411    Diesel       Manual      First  20.54 kmpl
7249              59000    Petrol    Automatic      First  17.21 kmpl
7250              28000    Diesel       Manual      First  23.08 kmpl
7251              52262    Petrol    Automatic      Third   17.2 kmpl
7252              72443    Diesel    Automatic      First   10.0 kmpl

      Engine      Power  Seats  New_Price  Price  Car_Age          Brand  \
0     998 CC  58.16 bhp    5.0        NaN   1.75       16         Maruti
1    1582 CC  126.2 bhp    5.0        NaN  12.50       11        Hyundai
2    1199 CC   88.7 bhp    5.0  8.61 Lakh   4.50       15          Honda
3    1248 CC  88.76 bhp    7.0        NaN   6.00       14         Maruti
4    1968 CC  140.8 bhp    5.0        NaN  17.74       13           Audi
...      ...        ...    ...        ...    ...      ...            ...
7248 1598 CC  103.6 bhp    5.0        NaN    NaN       15     Volkswagen
7249 1197 CC  103.6 bhp    5.0        NaN    NaN       11     Volkswagen
7250 1461 CC   63.1 bhp    5.0        NaN    NaN       14         Nissan
7251 1197 CC  103.6 bhp    5.0        NaN    NaN       13     Volkswagen
7252 2148 CC    170 bhp    5.0        NaN    NaN       12  Mercedes-Benz

               Model
0             WagonR
1           Creta1.6
2              JazzV
3          ErtigaVDI
4               A4New
...               ...
7248      VentoDiesel
7249           PoloGT
7250      MicraDiesel
7251           PoloGT
7252  E-Class2009-2013

[7252 rows x 16 columns]
```

Shape, Columns & Memory Analysis

[ ]

```
# Rows and columns
data.shape
```

```
(7252, 16)
```

[ ]

```
# Column names
data.columns
```

```
Index(['Name', 'Location', 'Year', 'Kilometers_Driven', 'Fuel_Type',
       'Transmission', 'Owner_Type', 'Mileage', 'Engine', 'Power', 'Seats',
       'New_Price', 'Price', 'Car_Age', 'Brand', 'Model'],
      dtype='object')
```

[ ]

```
# Memory usage
data.memory_usage(deep=True)
```

	0
Index	58016

Name	545050
Location	405004
Year	58016
Kilometers_Driven	58016
Fuel_Type	398642
Transmission	405007
Owner_Type	392868
Mileage	423443
Engine	404292
Power	411749
Seats	58016
New_Price	258642
Price	58016
Car_Age	58016
Brand	402485
Model	417684

dtype: int64

This command shows how much RAM (memory) each column of your DataFrame is using.

It returns memory in bytes.

2. Sorting & Top Records Exploration

```
# Top expensive cars
data.sort_values(by='Price', ascending=False).head(10)
```

	Name	Location	Year	Kilometers_Driven	Fuel_Type	Transmission	Owner_Type	Mileage	Engine	Power	Seats	New_Price	Price	Car_Age	Brand	Model
4079	Land Rover Range Rover 3.0 Diesel LWB Vogue	Hyderabad	2017	25000	Diesel	Automatic	First	13.33 kmpl	2993 CC	255 bhp	5.0	2.3 Cr	160.00	9	Land Rover	RoverRange
5781	Lamborghini Gallardo Coupe	Delhi	2011	6500	Petrol	Automatic	Third	6.4 kmpl	5204 CC	560 bhp	2.0	NaN	120.00	15	Lamborghini	GallardoCoupe
5919	Jaguar F Type 5.0 V8 S	Hyderabad	2015	8000	Petrol	Automatic	First	12.5 kmpl	5000 CC	488.1 bhp	2.0	NaN	100.00	11	Jaguar	FType
1505	Land Rover Range Rover Sport SE	Kochi	2019	26013	Diesel	Automatic	First	12.65 kmpl	2993 CC	255 bhp	5.0	1.39 Cr	97.07	7	Land Rover	RoverRange
1974	BMW 7 Series 740Li	Coimbatore	2018	28060	Petrol	Automatic	First	12.05 kmpl	2979 CC	320 bhp	5.0	NaN	93.67	8	BMW	7Series
1984	BMW 7 Series 740Li	Bangalore	2017	17465	Petrol	Automatic	First	12.05 kmpl	2979 CC	320 bhp	5.0	NaN	93.00	9	BMW	7Series
4691	Mercedes-Benz SLK-Class 55 AMG	Bangalore	2014	3000	Petrol	Automatic	Second	12.0 kmpl	5461 CC	421 bhp	2.0	NaN	90.00	12	Mercedes-Benz	SLK-Class55
5535	BMW X6 xDrive 40d M Sport	Ahmedabad	2015	97003	Diesel	Automatic	First	15.87 kmpl	2993 CC	308.43 bhp	5.0	NaN	85.00	11	BMW	X6xDrive
2095	Mercedes-Benz SLC 43 AMG	Coimbatore	2019	2526	Petrol	Automatic	First	19.0 kmpl	2996 CC	362.07 bhp	2.0	1.06 Cr	83.96	7	Mercedes-Benz	SLC43
2422	Jaguar XJ 3.0L Portfolio	Delhi	2016	12000	Diesel	Automatic	First	10.5 kmpl	2993 CC	271.23 bhp	4.0	NaN	79.00	10	Jaguar	XJ3.0L

```
# Oldest cars
data.sort_values(by='Car_Age', ascending=False).head()
```

	Name	Location	Year	Kilometers_Driven	Fuel_Type	Transmission	Owner_Type	Mileage	Engine	Power	Seats	New_Price	Price	Car_Age	Brand	Model
6216	Hindustan Motors Contessa 2.0 DSL	Pune	1996	65000	Diesel	Manual	Second	14.1 kmpl	1995 CC	null bhp	5.0	NaN	NaN	30	Hindustan	MotorsContessa
4709	Maruti 1000 AC	Hyderabad	1998	104000	Petrol	Manual	Second	15.0 kmpl	970 CC	null bhp	5.0	NaN	0.85	28	Maruti	1000AC
3749	Mercedes-Benz E-Class 250 D W 210	Mumbai	1998	55300	Diesel	Automatic	First	10.0 kmpl	1796 CC	157.7 bhp	5.0	NaN	3.90	28	Mercedes-Benz	E-Class250
3138	Maruti Zen LXI	Jaipur	1998	95150	Petrol	Manual	Third	17.3 kmpl	993 CC	60 bhp	5.0	NaN	0.45	28	Maruti	ZenLXI
5716	Maruti Zen LX	Jaipur	1998	95150	Petrol	Manual	Third	17.3 kmpl	993 CC	60 bhp	5.0	NaN	0.53	28	Maruti	ZenLX

Filtering / Query Operations

```
# Petrol cars only
data[data['Fuel_Type'] == 'Petrol']
```

	Name	Location	Year	Kilometers_Driven	Fuel_Type	Transmission	Owner_Type	Mileage	Engine	Power	Seats	New_Price	Price	Car_Age	Brand	Model

2	Honda Jazz V	Chennai	2011	46000	Petrol	Manual	First	18.2 kmpl	1199 CC	88.7 bhp	5.0	8.61 Lakh	4.50	15	Honda	JazzV
10	Maruti Ciaz Zeta	Kochi	2018	25692	Petrol	Manual	First	21.56 kmpl	1462 CC	103.25 bhp	5.0	10.65 Lakh	9.95	8	Maruti	CiazZeta
11	Honda City 1.5 V AT Sunroof	Kolkata	2012	60000	Petrol	Automatic	First	16.8 kmpl	1497 CC	116.3 bhp	5.0	NaN	4.49	14	Honda	City1.5
22	Audi A6 2011-2015 35 TFSI Technology	Mumbai	2015	55985	Petrol	Automatic	First	13.53 kmpl	1984 CC	177.01 bhp	5.0	NaN	23.50	11	Audi	A62011-2015
23	Hyundai i20 1.2 Magna	Kolkata	2010	45807	Petrol	Manual	First	18.5 kmpl	1197 CC	80 bhp	5.0	NaN	1.87	16	Hyundai	i201.2
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
7245	Honda Amaze S i-Vtech	Kochi	2015	44776	Petrol	Manual	First	18.0 kmpl	1198 CC	86.7 bhp	5.0	NaN	NaN	11	Honda	AmazeS
7246	Hyundai Grand i10 AT Asta	Coimbatore	2016	18242	Petrol	Automatic	First	18.9 kmpl	1197 CC	82 bhp	5.0	NaN	NaN	10	Hyundai	Grandi10
7247	Hyundai EON D Lite Plus	Coimbatore	2015	21190	Petrol	Manual	First	21.1 kmpl	814 CC	55.2 bhp	5.0	NaN	NaN	11	Hyundai	EOND
7249	Volkswagen Polo GT TSI	Mumbai	2015	59000	Petrol	Automatic	First	17.21 kmpl	1197 CC	103.6 bhp	5.0	NaN	NaN	11	Volkswagen	PoloGT
7251	Volkswagen Polo GT TSI	Pune	2013	52262	Petrol	Automatic	Third	17.2 kmpl	1197 CC	103.6 bhp	5.0	NaN	NaN	13	Volkswagen	PoloGT

3324 rows × 16 columns

```
# Multiple condition
data[(data['Car_Age'] < 5) & (data['Price'] > 10)]
```

Name	Location	Year	Kilometers_Driven	Fuel_Type	Transmission	Owner_Type	Mileage	Engine	Power	Seats	New_Price	Price	Car_Age	Brand	Model

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outlier detection methods

```
import pandas as pd
import numpy as np

data = pd.DataFrame({'Marks':[45,50,52,48,49,300]})
data
```

	Marks
0	45
1	50
2	52
3	48
4	49
5	300

```
Q1 = data['Marks'].quantile(0.25)
Q3 = data['Marks'].quantile(0.75)
IQR = Q3 - Q1

lower = Q1 - 1.5*IQR
upper = Q3 + 1.5*IQR

outliers_iqr = data[(data['Marks'] < lower) | (data['Marks'] > upper)]
outliers_iqr
```

```
import pandas as pd
import re
import matplotlib.pyplot as plt
from collections import Counter
from sklearn.feature_extraction.text import CountVectorizer, TfidfVectorizer
```

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FULL NLP EDA PIPELINE (Using Simple Text Data for Students)

```
import re
import nltk
import matplotlib.pyplot as plt

# Download NLTK resources (run once)
```

```
nltk.download('punkt')
nltk.download('stopwords')
nltk.download('wordnet')
nltk.download('averaged_perceptron_tagger')
```

```
[nltk_data] Downloading package punkt to /root/nltk_data...
[nltk_data]   Package punkt is already up-to-date!
[nltk_data] Downloading package stopwords to /root/nltk_data...
[nltk_data]   Package stopwords is already up-to-date!
[nltk_data] Downloading package wordnet to /root/nltk_data...
[nltk_data]   Package wordnet is already up-to-date!
[nltk_data] Downloading package averaged_perceptron_tagger to
[nltk_data]     /root/nltk_data...
[nltk_data]   Package averaged_perceptron_tagger is already up-to-
[nltk_data]       date!
True
```

```
from nltk.tokenize import word_tokenize
from nltk.corpus import stopwords
from nltk.stem import PorterStemmer, WordNetLemmatizer
from sklearn.feature_extraction.text import CountVectorizer, TfidfVectorizer
from collections import Counter
import pandas as pd
```

```
text_data = [
    "Machine Learning is transforming the world",
    "Students are learning Natural Language Processing",
    "Running models requires clean data",
    "Deep learning models are powerful and fast",
    "Happy students enjoy coding and learning"
]

data = pd.DataFrame({'text': text_data})
```

==========

1. TEXT CLEANING

==========

```
def clean_text(text):
    text = text.lower()
    text = re.sub(r'[^a-zA-Z\s]', '', text)
    return text

data['clean_text'] = data['text'].apply(clean_text)
```

```
data['clean_text']
```

	clean_text
0	machine learning is transforming the world
1	students are learning natural language processing
2	running models requires clean data
3	deep learning models are powerful and fast
4	happy students enjoy coding and learning

dtype: object

==========

2. TOKENIZATION

```
# ==========
data['tokens'] = data['clean_text'].apply(word_tokenize)
```

```
data['tokens']
```

	tokens
0	[machine, learning, is, transforming, the, world]
1	[students, are, learning, natural, language, p...
2	[running, models, requires, clean, data]
3	[deep, learning, models, are, powerful, and, f...
4	[happy, students, enjoy, coding, and, learning]

dtype: object

3. STOPWORD REMOVAL

==========

```
stop_words = set(stopwords.words('english'))

data['tokens'] = data['tokens'].apply(
    lambda x: [w for w in x if w not in stop_words]
)
```

```
data['tokens']
```

	tokens
0	[machine, learning, transforming, world]
1	[students, learning, natural, language, proces...
2	[running, models, requires, clean, data]
3	[deep, learning, models, powerful, fast]
4	[happy, students, enjoy, coding, learning]

dtype: object

4. POS TAGGING

```
# ========================================

# ========================================
data['pos_tags'] = data['tokens'].apply(nltk.pos_tag)

# SHOW FINAL OUTPUT
print("\nFinal NLP EDA Output:\n")
print(data[['text','tokens','pos_tags']])
```

```
Final NLP EDA Output:

                                    text  \
0        Machine Learning is transforming the world
1  Students are learning Natural Language Processing
2               Running models requires clean data
3        Deep learning models are powerful and fast
4          Happy students enjoy coding and learning

                                   tokens  \
0          [machine, learning, transforming, world]
1  [students, learning, natural, language, proces...
2          [running, models, requires, clean, data]
3          [deep, learning, models, powerful, fast]
4        [happy, students, enjoy, coding, learning]

                                    pos_tags
0  [(machine, NN), (learning, VBG), (transforming...
1  [(students, NNS), (learning, VBG), (natural, J...
2  [(running, VBG), (models, NNS), (requires, VBZ...
3  [(deep, JJ), (learning, NN), (models, NNS), (p...
4  [(happy, JJ), (students, NNS), (enjoy, VBP), (...
```

Double-click (or enter) to edit

```
# ================================
file_path = "used_cars_data.csv"
data = pd.read_csv(file_path)
```

```
print("Dataset Shape:", data.shape)
print("\nColumns:\n", data.columns.tolist())
```

```
Dataset Shape: (7253, 14)

Columns:
 ['S.No.', 'Name', 'Location', 'Year', 'Kilometers_Driven', 'Fuel_Type', 'Transmission', 'Owner_Type', 'Mileage', 'Engine', 'Power', 'Seats', 'New_Price', 'Price']
```

Double-click (or enter) to edit

```
# TEXT COLUMN
data['text'] = data['Name'].astype(str)
```

Double-click (or enter) to edit

2. Select text column (object type)

```
text_cols = data.select_dtypes(include="object").columns.tolist()
if len(text_cols) == 0:
    raise ValueError("No text column found in dataset for NLP EDA")
```

```
text_column = text_cols[0]
print("\nUsing text column for NLP EDA:", text_column)

data[text_column] = data[text_column].astype(str).fillna("")
```

```
Using text column for NLP EDA: Name
```

✅ 1. Text Cleaning (Basic NLP EDA)

Before stemming or lemmatization, clean the text.

```
def clean_text(text):
    text = text.lower()
    text = re.sub(r'[^a-zA-Z\s]', '', text)
    return text

data['clean_text'] = data['text'].apply(clean_text)
```

```
data['clean_text']
```

	clean_text
0	maruti wagon r lxi cng
1	hyundai creta crdi sx option
2	honda jazz v
3	maruti ertiga vdi
4	audi a new tdi multitronic
...	...
7248	volkswagen vento diesel trendline
7249	volkswagen polo gt tsi
7250	nissan micra diesel xv
7251	volkswagen polo gt tsi
7252	mercedesbenz eclass e cdi avantgarde

7253 rows × 1 columns

dtype: object

2. TOKENIZATION

```
import nltk
# Downloads (run once)
nltk.download('punkt')
nltk.download('stopwords')
nltk.download('wordnet')
nltk.download('averaged_perceptron_tagger')
nltk.download('punkt_tab')

from nltk.tokenize import word_tokenize
from nltk.corpus import stopwords
from nltk.stem import PorterStemmer, WordNetLemmatizer
from sklearn.feature_extraction.text import CountVectorizer, TfidfVectorizer
from collections import Counter
```

```
[nltk_data] Downloading package punkt to /root/nltk_data...
[nltk_data]   Package punkt is already up-to-date!
[nltk_data] Downloading package stopwords to /root/nltk_data...
[nltk_data]   Package stopwords is already up-to-date!
[nltk_data] Downloading package wordnet to /root/nltk_data...
[nltk_data]   Package wordnet is already up-to-date!
[nltk_data] Downloading package averaged_perceptron_tagger to
[nltk_data]     /root/nltk_data...
[nltk_data]   Package averaged_perceptron_tagger is already up-to-
[nltk_data]       date!
[nltk_data] Downloading package punkt_tab to /root/nltk_data...
[nltk_data]   Unzipping tokenizers/punkt_tab.zip.
```

Double-click (or enter) to edit

```
data['tokens'] = data['clean_text'].apply(word_tokenize)
```

```
data['tokens']
```

	tokens
0	[maruti, wagon, r, lxi, cng]
1	[hyundai, creta, crdi, sx, option]

2	[honda, jazz, v]
3	[maruti, ertiga, vdi]
4	[audi, a, new, tdi, multitronic]
...	...
7248	[volkswagen, vento, diesel, trendline]
7249	[volkswagen, polo, gt, tsi]
7250	[nissan, micra, diesel, xv]
7251	[volkswagen, polo, gt, tsi]
7252	[mercedesbenz, eclass, e, cdi, avantgarde]

7253 rows × 1 columns

dtype: object

3. STOPWORD REMOVAL

```
# STOPWORDS
from nltk.corpus import stopwords
stop_words = set(stopwords.words('english'))

# VERY IMPORTANT: lowercase comparison
data['tokens'] = data['tokens'].apply(
    lambda words: [w.lower() for w in words if w.lower() not in stop_words]
)
```

```
data['tokens']
```

	tokens
0	[machine, learning, transforming, world]
1	[students, learning, natural, language, proces...
2	[running, models, requires, clean, data]
3	[deep, learning, models, powerful, fast]
4	[happy, students, enjoy, coding, learning]

dtype: object

==========

4. STEMMING

==========

```
stemmer = PorterStemmer()

data['stemmed'] = data['tokens'].apply(
    lambda words: [stemmer.stem(w) for w in words]
)
```

```
data['stemmed']
```

	stemmed
0	[maruti, wagon, r, lxi, cng]
1	[hyundai, creta, crdi, sx, option]
2	[honda, jazz, v]
3	[maruti, ertiga, vdi]
4	[audi, new, tdi, multitron]
...	...
7248	[volkswagen, vento, diesel, trendlin]
7249	[volkswagen, polo, gt, tsi]
7250	[nissan, micra, diesel, xv]
7251	[volkswagen, polo, gt, tsi]
7252	[mercedesbenz, eclass, e, cdi, avantgard]

7253 rows × 1 columns

dtype: object

==========

5. LEMMATIZATION

==========

```
lemmatizer = WordNetLemmatizer()
```

```
data['lemmatized'] = data['tokens'].apply(
    lambda words: [lemmatizer.lemmatize(w) for w in words]
)
```

```
data['lemmatized']
```

	lemmatized
0	[maruti, wagon, r, lxi, cng]
1	[hyundai, creta, crdi, sx, option]
2	[honda, jazz, v]
3	[maruti, ertiga, vdi]
4	[audi, new, tdi, multitronic]
...	...
7248	[volkswagen, vento, diesel, trendline]
7249	[volkswagen, polo, gt, tsi]
7250	[nissan, micra, diesel, xv]
7251	[volkswagen, polo, gt, tsi]
7252	[mercedesbenz, eclass, e, cdi, avantgarde]

7253 rows × 1 columns

dtype: object

==========

6. WORD FREQUENCY ANALYSIS

```
# ==========================================
all_words = sum(data['lemmatized'], [])
print(Counter(all_words).most_common(20))

# ==========================================
```

```
[('maruti', 1444), ('hyundai', 1340), ('honda', 743), ('diesel', 609), ('toyota', 507), ('tdi', 479), ('mt', 427), ('swift', 418), ('mercedesbenz', 380), ('plus', 374), ('volkswagen', 374), ('vxi',
```

7. N-GRAM ANALYSIS

==========

```
vectorizer = CountVectorizer(ngram_range=(1,2))
X_ngram = vectorizer.fit_transform(data['clean_text'])

print("Ngram Shape:", X_ngram.shape)
```

```
Ngram Shape: (7253, 2595)
```

==========

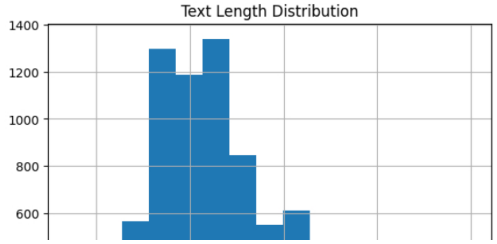
8. TEXT LENGTH VISUALIZATION

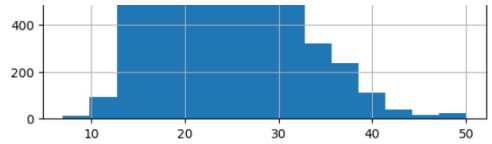
==========

```
import re
import nltk
import matplotlib.pyplot as plt
```

```
data['text_length'] = data['clean_text'].apply(len)

plt.figure()
data['text_length'].hist(bins=15)
plt.title("Text Length Distribution")
plt.show()
```





9. TF-IDF

==========

```
tfidf = TfidfVectorizer(max_features=1000)
X_tfidf = tfidf.fit_transform(data['clean_text'])

print("TFIDF Shape:", X_tfidf.shape)
```

```
TFIDF Shape: (7253, 631)
```

==========

10. POS TAGGING

==========

```
nltk.download('averaged_perceptron_tagger_eng')
```

```
[nltk_data] Downloading package averaged_perceptron_tagger_eng to
[nltk_data]     /root/nltk_data...
[nltk_data]   Unzipping taggers/averaged_perceptron_tagger_eng.zip.
True
```

```
data['pos_tags'] = data['tokens'].apply(nltk.pos_tag)

print(data[['text','tokens','stemmed','lemmatized','pos_tags']].head())
```

```
                            text                          tokens  \
0           Maruti Wagon R LXI CNG       [maruti, wagon, r, lxi, cng]
1  Hyundai Creta 1.6 CRDi SX Option  [hyundai, creta, crdi, sx, option]
2                    Honda Jazz V                   [honda, jazz, v]
3               Maruti Ertiga VDI              [maruti, ertiga, vdi]
4   Audi A4 New 2.0 TDI Multitronic      [audi, new, tdi, multitronic]

                          stemmed                        lemmatized  \
0       [maruti, wagon, r, lxi, cng]       [maruti, wagon, r, lxi, cng]
1  [hyundai, creta, crdi, sx, option]  [hyundai, creta, crdi, sx, option]
2                   [honda, jazz, v]                   [honda, jazz, v]
3              [maruti, ertiga, vdi]              [maruti, ertiga, vdi]
4        [audi, new, tdi, multitron]      [audi, new, tdi, multitronic]

                                    pos_tags
0  [(maruti, NN), (wagon, NN), (r, NN), (lxi, NN)...
1  [(hyundai, NN), (creta, NN), (crdi, NN), (sx, ...
2                [(honda, NN), (jazz, NN), (v, NN)]
3           [(maruti, NN), (ertiga, NN), (vdi, NN)]
4  [(audi, RB), (new, JJ), (tdi, NN), (multitroni...
```

2. Initial Exploration

What does this data look like?

This step gives you a snapshot of your data: how many rows and columns it has, what types of information are in it, and some basic statistics.

df.head() # Shows the first 5 rows

df.info() # Tells you data types and if ##anything is missing

df.describe() # Gives average, min, max, etc., for numbers

Why it matters:

Before doing any analysis, you need to understand what's in your data and if there are any problems, like missing or incorrect entries.

3. Handling Missing Values

What if there are gaps in the data?

Real-world data is often messy. This step is where we fix or remove missing values.

df.isnull().sum() # See how ##many values are missing

df['Age'].fillna(df['Age'].median()) # Fill ##missing age with the middle value

df.dropna() # Remove #rows with missing data

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